# Chapter 12. Quotients

*Cosets, quotient groups and spaces, isomorphism theorems*

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## 12.1 Cosets

**Motivation.** Cosets of $H$ partition $G$ into pieces of size $|H|$: Lagrange's theorem.

**Definition 12.1.1 (Cosets)**

For $H \leq G$, $x \in G$: $xH = \{xh : h \in H\}$ (or $x + H$ additively) is the **left coset** of $H$ with **representative** $x$, and $Hx$ the **right coset**. "Coset" means left coset from now on.

**Observation 12.1.2**

For $H \leq G$, $x, y \in G$: (a) $x \in xH$, so $G = \bigcup_{x \in G} xH$;
(b) $xH = yH$ iff $y^{-1}x \in H$;
(c) $xH \cap yH = \emptyset$ if $xH \neq yH$.

**Proof.** (a) $x = x1$. (b) If $xH = yH$, then $x = yh'$ by (a), so $y^{-1}x = h' \in H$. Conversely, if $y^{-1}x = h_0 \in H$, then $xh = yh_0h$ gives $xH \subseteq yH$, and $x^{-1}y = h_0^{-1}$ gives $yH \subseteq xH$. (c) If $xh_1 = yh_2$, then $y^{-1}x = h_2h_1^{-1} \in H$, so $xH = yH$ by (b). ■

So the cosets partition $G$ by the equivalence relation

$$x \equiv y \pmod{H} \iff xH = yH \iff y^{-1}x \in H,$$

with classes $\bar{x} = xH$. The set of cosets $G/H = \frac{G}{H} = \{xH : x \in G\} = \{\bar{x} : x \in G\}$ is the **coset space** of $G$ modulo $H$. For a **complete set of representatives** $\{x_i\}_{i \in I}$, $G/H = \{\bar{x}_i\}$ and $G = \coprod_i x_iH$ (**coset decomposition**).

**Example 12.1.3**

$1H = H$ is always a coset.

**Exercise 12.1.4**

$|G/G| = 1$ and $|G/\{1\}| = |G|$.

**Solution.** $xG = G$ for all $x$; $x\{1\} = \{x\}$ are all distinct. □

**Example 12.1.5**

The solution set $X_0 + \ker L_A$ of $AX = B$ is a coset of the subgroup $\ker L_A$ of the abelian group $(F^n, +)$.

**Example 12.1.6**

For $n\mathbb{Z} \leq \mathbb{Z}$ $(n \neq 0)$: $\bar{a} = a + n\mathbb{Z} = \bar{b} \iff a - b \in n\mathbb{Z} \iff n \mid (a - b)$, i.e. equal remainders mod $n$. So $\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \ldots, \overline{n-1}\}$; $\bar{a}$ is the **residue class** of $a$, and a complete set of representatives a **complete residue system**.

**Example 12.1.7**

In $S_3$ (notation of Example 11.8.4) with $H = \{1, \sigma\}$: $\tau H = \{\tau, \sigma\tau^2\}$, $\tau^2 H = \{\tau^2, \sigma\tau\}$, so $S_3/H = \{\bar{1}, \bar{\tau}, \overline{\tau^2}\}$. Since $\sigma\tau \neq \tau\sigma$, $\tau H \neq H\tau$: left and right cosets need not coincide.

**Exercise 12.1.8**

For $K = \{1, \tau, \tau^2\} \leq S_3$: $|S_3/K| = 2$, and left and right cosets coincide (e.g. $\sigma K = K\sigma$).

**Solution.** $S_3 = K \amalg \sigma K$ with $\sigma K = \{\sigma, \sigma\tau, \sigma\tau^2\}$, the three transpositions. Since $K\sigma$ is a coset of size 3 disjoint from $K$, it equals $\sigma K$. □

**Observation 12.1.9**

For $H \leq G$, $x, y \in G$: $|xH| = |yH| = |H|$.

**Proof.** $\lambda_x : H \to xH$, $h \mapsto xh$, is a bijection (injective by cancellation, onto by definition). ■

**Observation 12.1.10 (Lagrange's theorem)**

If $G$ is finite and $H \leq G$, then $|H|$ divides $|G|$ and $|G/H| = \frac{|G|}{|H|}$.

**Proof.** The distinct cosets partition $G$ and each has $|H|$ elements (Observation 12.1.9). ■

**Notation 12.1.11**

If $|G/H| < \infty$ ($G$ may be infinite), $[G : H] = |G/H|$ is the **index** of $H$ in $G$.

**Observation 12.1.12**

If $K \leq H \leq G$ with $|G/H|, |H/K| < \infty$, then $[G : K] = [G : H][H : K]$.

**Proof.** For representatives $\{g_i\}$ of $G/H$ and $\{h_j\}$ of $H/K$, $G = \coprod_i g_i H = \bigcup_{i,j} g_i h_j K$, disjointly: $g_i h_j K = g_{i'} h_{j'} K$ forces $g_i H = g_{i'} H$, so $i = i'$, then $h_j K = h_{j'} K$, so $j = j'$. ■

**Example 12.1.13**

By Example 11.3.6(e) and Lagrange, $\mu_m \leq \mu_n$ iff $m \mid n$.

**Observation 12.1.14**

If $G$ is finite and $x \in G$, then $|x|$ divides $|G|$.

**Proof.** $|x| = |\langle x \rangle|$; apply Lagrange. ■

**Proposition 12.1.15**

If $|G| = p$ is prime, then $G$ is cyclic, $G \approx \mu_p$.

**Proof.** Take $x \neq 1$. Then $|x| \neq 1$ divides $p$, so $|x| = p$, $\langle x \rangle = G$, and $\langle x \rangle \approx \mu_p$ (Exercise 11.7.9). ■

**Example 12.1.16**

Proper nontrivial subgroups of $S_3$ have order 2 or 3, hence are cyclic (Proposition 12.1.15): $\{id, (1\,2)\}$, $\{id, (2\,3)\}$, $\{id, (1\,3)\}$ and $\{id, (1\,2\,3), (1\,3\,2)\}$.

**Exercise 12.1.17**

If $|G|$ is even, $G$ has an element of order 2.

**Solution.** Pairing $x$ with $x^{-1}$ shows $\{x : x = x^{-1}\}$ has even size; it contains 1, hence some $x \neq 1$ with $|x| = 2$. □

**Remark 12.1.18**

But if $d \mid |G|$, $G$ need not have an element, or even a subgroup, of order $d$.

**Exercise 12.1.19**

(a) $S_4$ and $A_4$ have no elements of order 6, 8 or 12. (b) $S_4$ has exactly four subgroups of order 6. (c) $A_4$ has no subgroup of order 6. (d) How many subgroups of order 8 does $S_4$ have? (e) Of order 12?

**Solution.** (a) Elements of $S_4$ have cycle types $1, (2), (3), (4), (2,2)$, of orders $1, 2, 3, 4, 2$.

(b) The four stabilizers of a point ($\approx S_3$). Conversely, $|H| = 6$ gives a 3-cycle fixing some $i$ (Cauchy, Theorem 12.1.20); the order-2 elements of $H$ normalize its subgroup, so also fix $i$.

(c) Such $H$ would have index 2, hence be normal (Observation 12.2.9) and contain all squares, so all eight 3-cycles ($c = (c^2)^2$): too many.

(d) Three (the dihedral groups $D_4$, Sylow 2-subgroups).

(e) One, namely $A_4$ (a subgroup of index 2 contains all squares, which generate $A_4$). □

**Theorem 12.1.20 (Cauchy)**

If a prime $p$ divides $|G|$, then $G$ has an element of order $p$.

**Proof [Supplement] (McKay; the book defers it to abstract algebra).** Let $X = \{(x_1, \ldots, x_p) \in G^p : x_1 \cdots x_p = 1\}$; $|X| = |G|^{p-1} \equiv 0 \pmod p$. Since $x_2 \cdots x_p x_1 = x_1^{-1}(x_1 \cdots x_p)x_1 = 1$, cyclic shifts preserve $X$; their orbits have size $p$ or 1, the latter being $(x, \ldots, x)$ with $x^p = 1$. So $p$ divides $|\{x : x^p = 1\}| \geq 1$, which is thus $\geq p \geq 2$: some $x \neq 1$ has $|x| = p$. ■

## 12.2 Normal subgroups and quotient groups

**Motivation.** $\bar{x}\,\bar{y} = \overline{xy}$ is well defined exactly when the subgroup is normal.

For $\bar{x}\,\bar{y} = \overline{xy}$ on $G/H$ to be *well defined* we need: $x'^{-1}x, y'^{-1}y \in H \Rightarrow (x'y')^{-1}xy = y'^{-1}x'^{-1}xy \in H$.

**Example 12.2.1**

For $H = \{1, \sigma\} \leq S_3$ (Example 12.1.7): $\bar{1} = \bar{\sigma}$ and $\bar{\tau} = \overline{\sigma\tau^2}$, but $1 \cdot \tau = \tau$ while $\sigma \cdot \sigma\tau^2 = \tau^2$, and $\bar{\tau} \neq \overline{\tau^2}$. So the operation is not well defined. (For $K = \{1, \tau, \tau^2\}$ it is.)

**Definition 12.2.2 (Normal subgroup)**

$H \leq G$ is **normal**, $H \trianglelefteq G$, if $g^{-1}hg \in H$ for all $g \in G$, $h \in H$.

**Observation 12.2.3**

For $H \leq G$, $\bar{x}\,\bar{y} = \overline{xy}$ is well defined on $G/H$ iff $H \trianglelefteq G$.

**Proof.** If $H \trianglelefteq G$ and $x'^{-1}x, y'^{-1}y \in H$, then $y'^{-1}x'^{-1}xy = (y'^{-1}y)(y^{-1}(x'^{-1}x)y) \in H$. Conversely, if the operation is well defined, take $x = h$, $x' = 1$, $y = y' = g$: then $\bar{h} = \bar{1}$, so $\overline{hg} = \bar{g}$, i.e. $g^{-1}hg \in H$. ■

**Proposition 12.2.4**

If $N \trianglelefteq G$, then $G/N$ with $\bar{x}\,\bar{y} = \overline{xy}$ is a group, the **quotient group** of $G$ modulo $N$.

**Proof.** $(\bar{x}\bar{y})\bar{z} = \overline{xyz} = \bar{x}(\bar{y}\bar{z})$; $\bar{1}$ is the identity; $\bar{x}\,\overline{x^{-1}} = \bar{1} = \overline{x^{-1}}\,\bar{x}$, so $\bar{x}^{-1} = \overline{x^{-1}}$. ■

For $N \trianglelefteq G$: $ng = gn'$ with $n' = g^{-1}ng \in N$.

**Observation 12.2.5**

For $H \leq G$ the following are equivalent: (1) $H \trianglelefteq G$, i.e. $g^{-1}Hg \leq H$ for all $g$; (2) $gHg^{-1} = H$ for all $g$; (3) $g^{-1}Hg = H$ for all $g$.

**Proof.** (3)⇔(2): replace $g$ by $g^{-1}$. (3)⇒(1) is clear. (1)⇒(3): (1) for $g^{-1}$ gives $gHg^{-1} \leq H$, i.e. $H \leq g^{-1}Hg$. ■

**Exercise 12.2.6**

For $H \leq G$ the following are equivalent: (1) $H \trianglelefteq G$; (2) $\varphi(H) = H$ for every inner automorphism $\varphi$; (3) $gH = Hg$ for all $g$.

**Solution.** (1)⇔(2) is Observation 12.2.5 with $\varphi = \mathrm{Int}(g)$. (3)⇔ $gHg^{-1} = H$ (multiply by $g^{-1}$ on the right). □

**Example 12.2.7**

$\{1\} \trianglelefteq G$ and $G \trianglelefteq G$; $G/\{1\} \approx G$ and $G/G \approx \{1\}$.

**Example 12.2.8**

(a) In a commutative group every subgroup is normal, and $G/N$ is commutative.
(b) If $G = \langle x \rangle$ is cyclic, so is $G/N = \langle \bar{x} \rangle$.

**Observation 12.2.9**

If $[G : H] = 2$, then $H \trianglelefteq G$.

**Proof.** $G = H \amalg xH$ for $x \notin H$. For $h \in H$ and $g \notin H$, $g = xh_1$ and $ghg^{-1} = xh_2x^{-1}$, $h_2 = h_1hh_1^{-1} \in H$. If $xh_2x^{-1} \notin H$, then $xh_2x^{-1} = xh_3$, $h_3 \in H$, so $x = h_3^{-1}h_2 \in H$: contradiction. ■

(For index $3, 4, \ldots$ this fails, e.g. Example 12.2.1.)

**Example 12.2.10**

Groups of order 6. By Cauchy take $|x| = 2$, $|y| = 3$; $N = \langle y \rangle$ has index 2, so $N \trianglelefteq G = \{1, y, y^2, x, xy, xy^2\}$ and $x^{-1}yx \in \{y, y^2\}$. If $x^{-1}yx = y$, $|xy| = 6$ and $G \approx \mu_6$; if $x^{-1}yx = y^2$, $G \approx S_3$ (Examples 11.8.4, 11.9.23).

**Observation 12.2.11**

If $\varphi : G \to H$ is a homomorphism, then $\ker \varphi \trianglelefteq G$.

**Proof [Supplement].** For $k \in \ker \varphi$, $\varphi(g^{-1}kg) = \varphi(g)^{-1}\varphi(g) = e$. ■

(But $\operatorname{im} \varphi$ need not be normal in $H$.)

**Definition 12.2.12 (Natural projection)**

For $N \trianglelefteq G$, $\pi : G \to G/N$, $\pi(x) = \bar{x}$, is the **natural projection**.

**Exercise 12.2.13**

(a) $\pi$ is an epimorphism.
(b) $\ker \pi = N$.
(c) For the natural embedding $\iota : N \to G$, $\pi \circ \iota = \mathbf{1}$.

**Solution.** (a) $\pi(xy) = \overline{xy} = \bar{x}\bar{y}$, onto by definition.

(b) $\bar{x} = \bar{1} \iff x \in N$.

(c) is (b). □

The quotient operation is the unique one making $\pi$ a homomorphism.

**Notation 12.2.14**

For $1 \leq n \in \mathbb{Z}$, $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$, with $\bar{a} + \bar{b} = \overline{a + b}$.

**Observation 12.2.15**

$\mathbb{Z}_n$ is cyclic, so $\mathbb{Z}_n \approx \mu_n$.

**Proof.** $\mathbb{Z}_n = \langle \bar{1} \rangle$ has order $n$; apply Exercise 11.7.9. ■

**Exercise 12.2.16**

(a) In $A_4$, $N = \{id, (1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3)\}$ satisfies $N \approx \mathbb{Z}_2 \times \mathbb{Z}_2$ and $N \trianglelefteq A_4$.
(b) $H = \{id, (1\,2)(3\,4)\}$ satisfies $H \trianglelefteq N$ and $N \trianglelefteq A_4$, but $H$ is not normal in $A_4$.

**Solution.** (a) The non-identity elements have order 2 and any two multiply to the third: the Klein 4-group. $N$ is the identity plus all elements of type $(2,2)$, and conjugation preserves cycle type, so $N$ is normal (even in $S_4$).

(b) $[N : H] = 2$, so $H \trianglelefteq N$; but $(1\,2\,3)^{-1}(1\,2)(3\,4)(1\,2\,3) = (1\,4)(2\,3) \notin H$. □

**Remark 12.2.17**

This $N$ underlies solving the quartic by radicals; $A_5$ has no such normal subgroup, hence no quintic formula (Galois theory).

**Definition 12.2.18 (Center)**

$Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}$ is the **center** of $G$.

**Exercise 12.2.19**

(a) $Z(G) \trianglelefteq G$.
(b) If $\varphi \in \mathrm{Aut}(G)$, then $\varphi(Z(G)) = Z(G)$.

**Solution.** (a) $Z(G)$ is a subgroup, and $g^{-1}zg = z$ for $z \in Z(G)$.

(b) If $z$ commutes with all $g$, $\varphi(z)$ commutes with all $\varphi(g)$, i.e. with $G$; likewise for $\varphi^{-1}$. □

**Exercise 12.2.20**

(a) $Z(\mathrm{GL}_n(F)) = \{cI : c \in F^\times\}$.
(b) $Z(\mathrm{SL}_n(F)) = \{cI : c \in F,\ c^n = 1\}$.

**Solution.** (a) Scalars are central. If $A$ commutes with $I + E_{ij}$ $(i \neq j)$, it commutes with $E_{ij}$; $AE_{ij} = E_{ij}A$ forces $A_{ki} = 0$ for $k \neq i$, $A_{jk} = 0$ for $k \neq j$, and $A_{ii} = A_{jj}$. So $A$ is scalar.

(b) The same argument works, since $I + E_{ij} \in \mathrm{SL}_n(F)$; and $cI \in \mathrm{SL}_n(F)$ iff $c^n = 1$. □

**Definition 12.2.21 (Projective linear groups)**

$\mathbf{PGL}_n(F) = \mathrm{GL}_n(F)/Z(\mathrm{GL}_n(F))$ and $\mathbf{PSL}_n(F) = \mathrm{SL}_n(F)/Z(\mathrm{SL}_n(F))$; similarly **PSO**, **PSU**, ....

GL, **PGL**, SL, **PSL**, **O**, **SO**, **PSO**, **U**, **SU**, **PSU** (and **Sp**, **PSp** of Chapter 13) are Weyl's **classical groups**.

## 12.3 Quotient spaces

**Motivation.** $V/W$ is a vector space of dimension $\dim V - \dim W$, the tool for induction on dimension.

For $W \leq V$, $V/W$ is an abelian group with $\bar{u} + \bar{v} = \overline{u+v}$, $\bar{v} = v + W$, and $c\bar{v} = \overline{cv}$ is well defined: $v - u \in W$ gives $c(v-u) \in W$.

**Proposition 12.3.1**

For $W \leq V$, $V/W$ with $\bar{u} + \bar{v} = \overline{u+v}$ and $c\bar{v} = \overline{cv}$ is a vector space, the **quotient space** of $V$ modulo $W$.

**Proof.** Each axiom follows from the one in $V$ by putting bars on; zero $\bar{0}$, $-\bar{v} = \overline{-v}$. ■

**Example 12.3.2**

$V/0 \approx V$ and $V/V \approx 0$.

**Exercise 12.3.3**

For $W \leq V$: (a) $\pi : V \to V/W$ is a linear epimorphism;
(b) $\ker \pi = W$;
(c) for the inclusion $\iota : W \to V$, $\pi \circ \iota = 0$.

**Solution.** $\pi(cv + u) = \overline{cv+u} = c\bar{v} + \bar{u}$; onto by definition; $\bar{v} = \bar{0} \iff v \in W$. □

**Remark 12.3.4**

Heuristic: for $|F| = q$, Lagrange gives $|V/W| = q^n/q^m$, suggesting $\dim V/W = n - m$.

**Theorem 12.3.5**

If $V$ is finite-dimensional and $W \leq V$, then $\dim V = \dim W + \dim(V/W)$.

**Proof.** Apply the dimension theorem to $\pi : V \to V/W$ ($\ker \pi = W$, $\operatorname{im} \pi = V/W$). ■

**Exercise 12.3.6**

If a basis $\{v_1, \ldots, v_m\}$ of $W$ is extended to a basis $\{v_1, \ldots, v_n\}$ of $V$, then $\{\overline{v_{m+1}}, \ldots, \overline{v_n}\}$ is a basis of $V/W$.

**Solution.** Spanning: $\bar{v} = \overline{\sum a_i v_i} = \sum_{i>m} a_i \bar{v}_i$. Independence: $\sum_{i>m} a_i \bar{v}_i = \bar{0}$ gives $\sum_{i>m} a_i v_i \in \langle v_1, \ldots, v_m \rangle$, so all $a_i = 0$. □

**Observation 12.3.7**

Every subspace of $F^n$ is the solution space of $AX = 0$ for some $A \in \mathfrak{M}_{m,n}(F)$.

**Proof.** Take $A$ the matrix of $\pi : F^n \to F^n/W$ in some bases; $\ker \pi = W$. ■

## 12.4 Isomorphism theorems

**Motivation.** Dividing out its kernel turns a homomorphism into an isomorphism onto its image.

**Observation 12.4.1**

If $\varphi : G \to H$ is a homomorphism and $\varphi(g) = h$, then $\varphi^{-1}(h) = g \ker \varphi$.

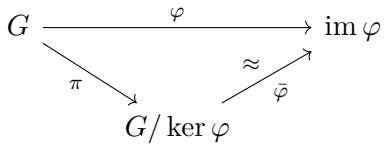
**Proof.** $x \in \varphi^{-1}(h) \iff \varphi(x) = \varphi(g) \iff g^{-1}x \in \ker\varphi \iff x \in g\ker\varphi$. ■

So the fibres of $\varphi$ are the cosets of $\ker\varphi$.

**Theorem 12.4.2 (First isomorphism theorem for groups)**

For a homomorphism $\varphi : G \to H$, $\bar\varphi : G/\ker\varphi \to \operatorname{im}\varphi$, $\bar\varphi(\bar g) = \varphi(g)$, is a well-defined isomorphism: $G/\ker\varphi \approx \operatorname{im}\varphi$.

**Proof.** Well defined: $g'^{-1}g \in \ker\varphi$ gives $\varphi(g) = \varphi(g')$. Homomorphism: $\bar\varphi(\bar g\bar g') = \varphi(gg') = \varphi(g)\varphi(g')$. Onto: clear. Injective: $\bar\varphi(\bar g) = 1$ gives $g \in \ker\varphi$, $\bar g = \bar 1$ (Observation 11.6.7). ■




**Remark 12.4.3**

(a) Writing $\bar G = G/\ker\varphi$, $\bar\varphi\bar g = \varphi g$. (b) Equivalently, $\psi : \operatorname{im}\varphi \to G/\ker\varphi$, $\psi(\varphi(g)) = \bar g$, is an isomorphism; its well-definedness ($\varphi(g) = \varphi(g') \Rightarrow \bar g = \bar g'$) is the injectivity of $\bar\varphi$.

**Example 12.4.4**

$\operatorname{sgn} : S_n \to \mu_2$ ($n \geq 2$) is onto with kernel $A_n$; so $A_n \trianglelefteq S_n$, $S_n/A_n \approx \mu_2$, $[S_n : A_n] = 2$, $S_n = A_n \amalg \tau A_n$ ($\tau \notin A_n$), and $|A_n| = n!/2$.

**Example 12.4.5**

$\det : \mathrm{GL}_n(F) \to F^\times$ is onto with kernel $\mathrm{SL}_n(F)$; so $\mathrm{SL}_n(F) \trianglelefteq \mathrm{GL}_n(F)$ and $\mathrm{GL}_n(F)/\mathrm{SL}_n(F) \approx F^\times$.

**Example 12.4.6**

(a) $\det : \mathbf{O}(n) \to \mu_2$ is onto with kernel $\mathbf{SO}(n)$; so $\mathbf{SO}(n) \trianglelefteq \mathbf{O}(n)$, $\mathbf{O}(n)/\mathbf{SO}(n) \approx \mu_2$, and $\mathbf{O}(n) = \mathbf{SO}(n) \amalg S \cdot \mathbf{SO}(n)$ ($S \notin \mathbf{SO}(n)$; Exercise 9.3.11).
(b) $\det : \mathbf{U}(n) \to S^1$ is onto (e.g. $\operatorname{diag}(z, 1, \dots, 1)$) with kernel $\mathbf{SU}(n)$; so $\mathbf{U}(n)/\mathbf{SU}(n) \approx S^1 \approx \mathbf{SO}(2)$.

**Exercise 12.4.7**

(a) For $A, B \in \mathrm{GL}_n(F)$, $\bar A = \bar B$ in $\mathrm{GL}_n(F)/\mathrm{SL}_n(F)$ iff $\det A = \det B$. (b) Describe the coset decomposition of $\mathrm{GL}_n(F)$ modulo $\mathrm{SL}_n(F)$. (c) Describe that of $\mathbf{U}(n)$ modulo $\mathbf{SU}(n)$.

**Solution.** (a) $\bar A = \bar B \iff B^{-1}A \in \mathrm{SL}_n(F) \iff \det A = \det B$. (b) $\mathrm{GL}_n(F) = \coprod_{c\in F^\times} \operatorname{diag}(c, 1, \dots, 1)\,\mathrm{SL}_n(F)$: one coset for each value of the determinant. (c) $\mathbf{U}(n) = \coprod_{z\in S^1} \operatorname{diag}(z, 1, \dots, 1)\mathbf{SU}(n)$. □

**Exercise 12.4.8**

$\rho : (\mathbb{R}, +) \to \mathbf{SO}(2)$, $\rho(\theta) = R_\theta$, is a homomorphism, and $\mathbb{R}/2\pi\mathbb{Z} \approx \mathbf{SO}(2) \approx S^1$.

**Solution.** $R_\theta R_\eta = R_{\theta+\eta}$; $\rho$ is onto (Theorem 9.5.2) with kernel $2\pi\mathbb{Z}$; apply Theorem 12.4.2 and Example 11.9.19. □

**Example 12.4.9**

For $x \in G$, $\gamma_x : \mathbb{Z} \to G$, $\gamma_x(a) = x^a$, has $\ker \gamma_x = n\mathbb{Z}$ for some $n \geq 0$ (Theorem 11.4.2) and image $\langle x \rangle$, so $\langle x \rangle \approx \mathbb{Z}/n\mathbb{Z} = \mathbb{Z}_n$ (with $\mathbb{Z}_0 = \mathbb{Z}$, $\mathbb{Z}_1 = 0$).

**Theorem 12.4.10**

Up to isomorphism the cyclic groups are exactly the $\mathbb{Z}_n$, $n \geq 0$.

**Proof.** Example 12.4.9 with $G = \langle x \rangle$; and $\mathbb{Z}_n$ is cyclic. ■

**Exercise 12.4.11**

For $\mathrm{Int} : G \to \mathrm{Aut}(G)$, $g \mapsto \mathrm{Int}(g)$: (a) Int is a homomorphism;
(b) $\mathrm{Int}(G) \trianglelefteq \mathrm{Aut}(G)$;
(c) $\ker(\mathrm{Int}) = Z(G)$, so $\mathrm{Int}(G) \approx G/Z(G)$.

**Solution.** (a) $\mathrm{Int}(gh)(x) = ghxh^{-1}g^{-1} = \mathrm{Int}(g)\,\mathrm{Int}(h)(x)$ (this is why $\mathrm{Int}(g)(x) = gxg^{-1}$ rather than $g^{-1}xg$). (b) For $\varphi \in \mathrm{Aut}(G)$: $\varphi \circ \mathrm{Int}(g) \circ \varphi^{-1}(x) = \varphi(g\varphi^{-1}(x)g^{-1}) = \varphi(g)x\varphi(g)^{-1}$, i.e. $= \mathrm{Int}(\varphi(g))$. (c) $\mathrm{Int}(g) = id \iff gx = xg$ for all $x$; apply Theorem 12.4.2. □

**Exercise 12.4.12**

Define $\psi : \mathrm{SL}_2(F) \to \mathrm{GL}_3(F)$ by

$$\psi\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} a^2 & -2ac & -c^2 \\ -ab & ad+bc & cd \\ -b^2 & 2bd & d^2 \end{pmatrix}.$$

(a) $\mathrm{im}\,\psi \leq \mathrm{SL}_3(F)$.
(b) $\psi$ is a homomorphism with $\ker \psi = \{\pm I\} = Z(\mathrm{SL}_2(F))$.
(c) Hence $\mathbf{PSL}_2(F) \approx \mathrm{im}\,\psi$ is a matrix group.

**Solution.** [Supplement] $\psi(A)$ is the action of $A$ on binary quadratic forms; multiplying out gives $\psi(AB) = \psi(A)\psi(B)$ and $\det \psi(A) = (ad - bc)^3$. (a) For $A \in \mathrm{SL}_2(F)$, $\det \psi(A) = 1^3 = 1$. (b) $\psi(A) = I$ forces $c = b = 0$ (off-diagonal entries $-c^2 = -b^2 = 0$) and $a^2 = d^2 = ad = 1$, so $a = d = \pm 1$; conversely $\psi(\pm I) = I$. And $Z(\mathrm{SL}_2(F)) = \{\pm I\}$ (Exercise 12.2.20, $c^2 = 1$). (c) First isomorphism theorem. □

**Exercise 12.4.13**

Is there an epimorphism $S_3 \to \mathbb{Z}_3$?

**Solution.** No: its kernel would be a normal subgroup of order 2, and there is none (Example 12.2.1). □

**Theorem 12.4.14 (First isomorphism theorem for vector spaces)**

For a linear map $L : V \to W$, $\bar{L} : V/\ker L \to \operatorname{im} L$, $\bar{L}\bar{v} = Lv$, is a well-defined vector space isomorphism.

**Proof.** By Theorem 12.4.2 for $L : (V, +) \to (W, +)$, $\bar{L}$ is a well-defined group isomorphism; and $\bar{L}(c\bar{v}) = \bar{L}(\overline{cv}) = L(cv) = cL(v) = c\bar{L}(\bar{v})$. ■

**Remark 12.4.15**

By Theorem 12.4.14, the dimension theorem and Theorem 12.3.5 are equivalent statements.

**Exercise 12.4.16**

For $\operatorname{tr} : \mathfrak{M}_{n,n}(F) \to F$: (a) $\ker \operatorname{tr} = \mathfrak{sl}_n(F)$;
(b) $\mathfrak{M}_{n,n}(F)/\mathfrak{sl}_n(F) \approx F$, so $\dim \mathfrak{sl}_n(F) = n^2 - 1$.

**Solution.** tr is linear and onto ($\operatorname{tr}(cE_{11}) = c$); $\mathfrak{sl}_n(F)$ is by definition the trace-zero matrices; apply Theorem 12.4.14. □

**Exercise 12.4.17**

For $T \in \mathfrak{L}(V, V)$ and a $T$-cyclic space $V = F[t]v$, $\varphi : F[t] \to V$, $\varphi(f) = f(T)v$: (a) $\varphi$ is linear with $\ker \varphi = \mathcal{I}_T$ (the annihilator ideal of $v$);
(b) $V \approx F[t]/\mathcal{I}_T$.

**Solution.** $\varphi$ is linear and onto by definition of $F[t]v$; its kernel is $\{f : f(T)v = 0\}$; apply Theorem 12.4.14. □

**Observation 12.4.18**

If $H \trianglelefteq G$ and $K \leq G$, then $HK \leq G$.

**Proof.** $1 \in HK$. For $h, h' \in H$, $k, k' \in K$: $(hk)(h'k') = h(kh'k^{-1})kk' \in HK$ and $(hk)^{-1} = k^{-1}h^{-1} = (k^{-1}h^{-1}k)k^{-1} \in HK$. ■

**Theorem 12.4.19 (Second isomorphism theorem for groups)**

If $H \trianglelefteq G$ and $K \leq G$, then $\bar{\jmath} : K/(H \cap K) \to HK/H$, $\bar{\jmath}(k(H \cap K)) = kH$, is a well-defined isomorphism.

**Proof.** By Observation 12.4.18 $HK \leq G$, and clearly $H \trianglelefteq HK$, $H \cap K \trianglelefteq K$.

(i) Well defined: if $k'^{-1}k \in H \cap K$, then $k'^{-1}k \in H$.

(ii) Injective: if $k'^{-1}k \in H$, then (as $k, k' \in K$) $k'^{-1}k \in H \cap K$.

(iii) Homomorphism: $\bar{\jmath}(\bar{k}\bar{k}') = \overline{kk'} = \bar{k}\,\bar{k}'$.

(iv) Onto: for $h \in H$, $k \in K$, $\overline{hk} = \bar{h}\,\bar{k} = \bar{1}\bar{k} = \bar{k}$ in $HK/H$ ($H$ normal). ■

**Second proof.** $j = \pi' \circ \iota : K \to HK \to HK/H$ is an epimorphism by (iv), with $\ker j = H \cap K$; apply Theorem 12.4.2. ■

**Corollary 12.4.20**

If $G$ is finite, $H \trianglelefteq G$, $K \leq G$, then $|HK| \cdot |H \cap K| = |H| \cdot |K|$.

**Proof.** Lagrange and Theorem 12.4.19: $\frac{|K|}{|H \cap K|} = \frac{|HK|}{|H|}$. ■

**Theorem 12.4.21 (Second isomorphism theorem for vector spaces)**

For $U, W \leq V$, $\bar{\jmath} : W/(U \cap W) \to (U + W)/U$, $\bar{\jmath}(\bar{w}) = \bar{w}$, is a well-defined vector space isomorphism.

**Proof.** Theorem 12.4.19 for $(V, +)$ gives a group isomorphism; it is $F$-linear since $\bar{\jmath}(c\bar{w}) = \overline{cw} = c\bar{w}$. ■

**Corollary 12.4.22**

For finite-dimensional $V$ and $U, W \leq V$: $\dim(U + W) + \dim(U \cap W) = \dim U + \dim W$.

**Proof.** By Theorems 12.4.21 and 12.3.5, $\dim W - \dim(U \cap W) = \dim(U + W) - \dim U$. ■

## 12.5 Triangularization II

**Motivation.** An invariant $W$ splits $T$ into $T|_W$ and $\bar{T}$ on $V/W$; induction on $\dim V$ triangularizes.

Spaces are finite-dimensional. For $T$-invariant $W$ with basis $\mathfrak{C} = \{v_1, \ldots, v_m\}$ extended to $\mathfrak{B} = \{v_1, \ldots, v_n\}$, $[T]^{\mathfrak{B}}_{\mathfrak{B}} = \begin{pmatrix} [T|_W]^{\mathfrak{C}}_{\mathfrak{C}} & * \\ 0 & ? \end{pmatrix}$; the block "?" is $\bar{T}\bar{v} = \overline{Tv}$ on $\bar{V} = V/W$ with basis $\bar{\mathfrak{B}} = \{\bar{v}_{m+1}, \ldots, \bar{v}_n\}$ (Exercise 12.3.6).

**Observation 12.5.1**

If $W \leq V$, $W' \leq V'$ and $L : V \to V'$ is linear with $LW \leq W'$, then $\bar{L} : V/W \to V'/W'$, $\bar{L}\bar{v} = \overline{Lv}$, is a well-defined linear map ($L$ **induces** $\bar{L}$).

**Proof.** If $v - u \in W$, then $Lv - Lu = L(v - u) \in W'$; linearity is inherited. (Conversely, well-definedness forces $LW \leq W'$: take $u = 0$.) ■

$$\begin{array}{ccc} V & \xrightarrow{L} & V' \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi} \\ V/W & \xrightarrow[\bar{L}]{} & V'/W' \end{array}$$

**Exercise 12.5.2**

State and prove Observation 12.5.1 for groups.

**Solution.** If $N \trianglelefteq G$, $N' \trianglelefteq G'$ and $\varphi : G \to G'$ is a homomorphism with $\varphi(N) \leq N'$, then $\bar{\varphi}(\bar{g}) = \overline{\varphi g}$ is a well-defined homomorphism $G/N \to G'/N'$: if $g'^{-1}g \in N$, then $\varphi(g')^{-1}\varphi(g) = \varphi(g'^{-1}g) \in N'$. □

**Exercise 12.5.3**

The well-definedness of $\bar{L}$ in Theorem 12.4.14 and of $\bar{\jmath}$ in Theorem 12.4.21 are special cases of Observation 12.5.1.

**Solution.** $\bar{L}$ is induced by $L : V \to \operatorname{im} L$ with $L(\ker L) \leq 0$; $\bar{\jmath}$ is induced by the inclusion $W \to U+W$ with $U \cap W \leq U$. □

**Lemma 12.5.4**

Let $W$ be $T$-invariant, $\bar{T} : \bar{V} \to \bar{V}$ as above, $\mathfrak{C} = \{v_1, \ldots, v_m\}$ a basis of $W$, $\mathfrak{B} = \{v_1, \ldots, v_n\}$ a basis of $V$, $\bar{\mathfrak{B}} = \{\bar{v}_{m+1}, \ldots, \bar{v}_n\}$. Then

$$[T]_{\mathfrak{B}}^{\mathfrak{B}} = \begin{pmatrix} [T|_W]_{\mathfrak{C}}^{\mathfrak{C}} & * \\ 0 & [\bar{T}]_{\bar{\mathfrak{B}}}^{\bar{\mathfrak{B}}} \end{pmatrix}.$$

**Proof.** The first $m$ columns are as stated since $TW \leq W$. From $Tv_{m+j} = \sum_i a_{i,m+j} v_i$ and $\bar{v}_1 = \cdots = \bar{v}_m = \bar{0}$, $\bar{T}\bar{v}_{m+j} = \sum_{i>m} a_{i,m+j}\bar{v}_i$: column $j$ of $[\bar{T}]$ is the lower part of column $m+j$. ■

**Exercise 12.5.5**

In Lemma 12.5.4: (a) $\operatorname{tr} T = \operatorname{tr}(T|_W) + \operatorname{tr}\bar{T}$;
(b) $\det T = \det(T|_W) \det \bar{T}$;
(c) $\phi_T(t) = \phi_{T|_W}(t)\phi_{\bar{T}}(t)$.

**Solution.** Read off the block-triangular matrix (for (c), apply the block determinant to $tI - [T]$). □

**Theorem 12.5.6 (Triangularization)**

If $F = \mathbb{C}$ and $T \in \mathfrak{L}(V, V)$, then $V$ has a basis $\mathfrak{B}$ with $[T]_{\mathfrak{B}}^{\mathfrak{B}}$ upper triangular.

**Second proof.** Induction on $n = \dim V$. Take an eigenvector $Tv = \lambda v$, $W = \langle v \rangle$, $\bar{V} = V/W$ of dimension $n-1$. By induction $\bar{V}$ has a basis $\bar{\mathfrak{D}} = \{\bar{u}_2, \ldots, \bar{u}_n\}$ with $[\bar{T}]_{\bar{\mathfrak{D}}}^{\bar{\mathfrak{D}}}$ upper triangular. $\mathfrak{D} = \{v, u_2, \ldots, u_n\}$ is a basis ($cv + \sum c_i u_i = 0$ gives $\sum c_i \bar{u}_i = 0$ under $\pi$, so $c_i = 0$, $c = 0$), and $[T]_{\mathfrak{D}}^{\mathfrak{D}} = \left(\begin{smallmatrix} \lambda & * \\ 0 & [\bar{T}]_{\bar{\mathfrak{D}}}^{\bar{\mathfrak{D}}} \end{smallmatrix}\right)$ (Lemma 12.5.4). ■

**Example 12.5.7**

Let $G$ be finite and $\varphi : G \to \mathrm{GL}_n(\mathbb{C})$ a homomorphism, $\chi = \operatorname{tr} \circ \varphi$ (the **character** of $\varphi$). Triangularize $\varphi(g)$ with diagonal $\lambda_1, \ldots, \lambda_n$; $\varphi(g)^{|g|} = I$ gives $\lambda_i^{|g|} = 1$, $|\lambda_i| = 1$, and $\varphi(g^{-1}) = \varphi(g)^{-1}$ has diagonal $\lambda_i^{-1} = \bar{\lambda}_i$, so

$$\chi(g^{-1}) = \sum \lambda_i^{-1} = \sum \bar{\lambda}_i = \overline{\chi(g)}.$$

**Exercise 12.5.8**

In Example 12.5.7, if $g$ is conjugate to $h$, then $\chi(g) = \chi(h)$.

**Solution.** $h = x^{-1}gx$ gives $\varphi(h) = \varphi(x)^{-1}\varphi(g)\varphi(x)$, and similar matrices have equal traces. □

**Definition 12.5.9 (Simultaneous triangularization)**

$T, S \in \mathfrak{L}(V, V)$ are **simultaneously triangularizable** if $[T]_{\mathfrak{B}}^{\mathfrak{B}}$ and $[S]_{\mathfrak{B}}^{\mathfrak{B}}$ are upper triangular for one basis $\mathfrak{B}$ (likewise for any family). Then the first vector of $\mathfrak{B}$ is a common eigenvector.

**Proposition 12.5.10**

If $F = \mathbb{C}$ and $TS = ST$, then $T$ and $S$ have a common eigenvector.

**Proof.** An eigenspace $E_\lambda^T$ is $S$-invariant ($T(Sv) = S(Tv) = \lambda Sv$); take an eigenvector of $S|_{E_\lambda^T}$. ■

**Proposition 12.5.11**

If $F = \mathbb{C}$ and $TS = ST$, then $T, S$ are simultaneously triangularizable.

**Proof.** For a common eigenvector $v$ (Proposition 12.5.10), $\bar{T}, \bar{S}$ on $V/\langle v \rangle$ commute; by induction they are simultaneously triangularizable, and lift as in Theorem 12.5.6. ■

**Exercise 12.5.12**

If $\{T_i\}_{i \in I} \subseteq \mathfrak{L}(V, V)$ ($F = \mathbb{C}$) pairwise commute, they are simultaneously triangularizable.

**Solution.**

*Step 1 (a common eigenvector).* Take a nonzero $U$ invariant under all $T_i$, of minimal dimension. If some $T_j|_U$ is not scalar, an eigenspace $\{u \in U : T_j u = \lambda u\}$ is a smaller such subspace (commutativity). So all $T_i$ are scalar on $U$, and any $0 \neq v \in U$ works.

*Step 2 (induction).* Pass to $V/\langle v \rangle$ as in Proposition 12.5.11. □